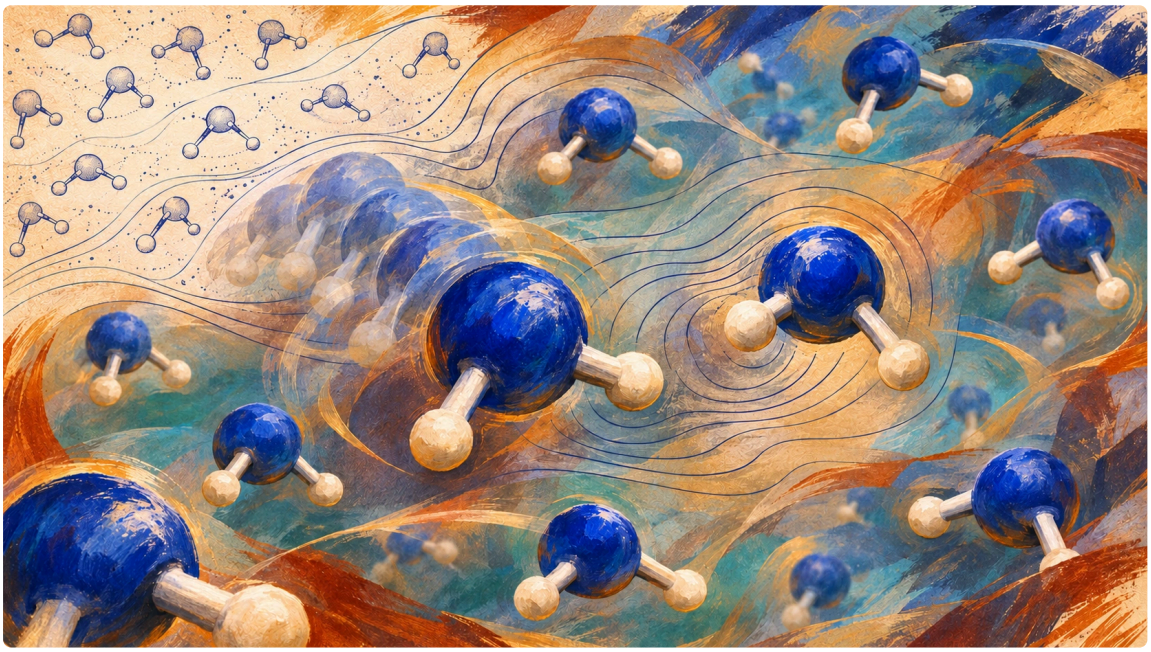
Machine-learning force fields: From interatomic forces to material properties

From the foundations of molecular dynamics to choices about model representations, training data, accuracy, and computational cost

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Samples of water configurations flow into continuous curves and overlapping views of molecular motion. This conceptual illustration connects learning an interaction function from configuration samples with calculating atomic motion. The curves and overlaps are not actual energy values or computed trajectories. AI-generated and edited.

1. Predicting materials through atomic motion

Which paths do ions follow through a battery? Which configurations does a drug molecule retain as it approaches a protein? Which bonds break first on a catalyst surface? Beneath these different questions lie the arrangements and motions of atoms. Calculating how atoms push and pull on one another lets us explore possible structures and pathways in processes that are difficult to observe directly.

Molecular dynamics (MD) calculates this motion. In 1964, Rahman simulated liquid argon with 864 atoms interacting through a Lennard-Jones potential. [1] The systems studied today extend to proteins, electrolytes, metals, and polymers. Then as now, advancing atomic positions requires the force on every atom in the current configuration. A force field is the interaction model that supplies those forces.

A machine-learning force field (MLFF) learns the relationship between atomic configurations and energies or forces from data. It seeks to represent complex environmental dependence while reducing the cost of repeating detailed electronic-structure calculations at every instant. To understand this possibility, we first need to follow what an energy function represents, how its gradient gives forces, and how repeated force calculations yield material properties. Model architecture, training data, and validation then take their place in that process.

A protein can share its environment with water, ions, lipids, and metal centers. A single interaction model covering these chemistries could also change how a simulation is assembled. SO3LR Tauri provides a case study. [21] After developing the MD foundations, we examine how it combines learning with explicit physical terms, and what has been validated at the level of observable properties.

2. Describing atomic configurations through interaction energies

Let $R=(R_1,\cdots,R_n)$ denote the coordinates of N atomic nuclei. The potential energy U(R) is a single number associated with the entire configuration. Moving the atoms changes this value. The function assigning an energy to each possible configuration is the potential-energy surface (PES). A drawing of a curved surface projects this function onto a few of its many coordinates. Each point represents a complete configuration, so its meaning differs from that of one atom rolling over a hill in physical space.

Molecular mechanics for biomolecules commonly separates the energy into contributions from bond lengths, bond angles, torsion angles, and nonbonded interactions. A representative expression is [2]

$$U(\mathbf{R}) = U_{\text{bond}} + U_{\text{angle}} + U_{\text{torsion}} + U_{\text{nonbonded}}$$

Term and example	Change described	Meaning of the parameters
Bond: $\frac{1}{2}k_\beta(r-r_0)^2$	Stretching the bond between two specified atoms	r_0 is the equilibrium length; k_β is a stiffness with units of energy/length².
Angle: $\frac{1}{2}k_a(\alpha-\alpha_0)^2$	Bending the angle formed by three atoms	α_0 is the equilibrium angle. The units of k_a must be consistent with angles expressed in radians.
Torsion: $\Sigma_n V_n[1+\cos(n\phi-\delta_n)]$	Rotational barriers as a function of the dihedral angle of four atoms	V_n sets the energy scale, n is the periodicity, and δ_n is the phase.
Nonbonded: $U^{LJ}(r_{ij})+k_e q_i q_j/r_{ij}$	Interatomic repulsion, dispersion, and electrostatics	q_i is a partial charge; k_e is the Coulomb constant in the chosen unit system. Pair contributions must be summed without indiscriminate double counting.

After choosing the functional form, parameters for atom types and chemical environments are fitted to quantum-chemical and experimental data. Even the harmonic expressions ½k(r−r₀)² and k′(r−r₀)² define their coefficients differently. Exclusions for bonded pairs, scaling of 1-4 nonbonded interactions, and combinations of water and ion models also belong to the chosen force field. Replacing or mixing parameter files therefore makes a physical choice that goes beyond changing a few symbols.

$$U^{\mathrm{LJ}}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

The Lennard-Jones (LJ) potential represents steep short-range repulsion and longer-range attraction with two terms. Here ε is the depth of the energy well and σ is the separation where the energy crosses zero. The minimum occurs at 2^(1/6)σ, where U=−ε. The r^{-6} term captures the long-range form of dispersion for suitable neutral atom pairs, whereas the r^{-12} repulsion is a convenient empirical approximation. This single pair function cannot describe directional covalent bonding in its entirety. [3]

Empirical models have long described the influence of the surrounding environment. EAM includes an embedding energy that depends on the density around a metal atom, while Tersoff changes the effective bond strength with neighboring atoms and bond angles. [4,5] Reactive and polarizable empirical force fields also exist. Presenting MLFFs as the first transition from simple pair interactions to complex physics would overlook these developments. The central change is the much greater flexibility with which interaction functions and their representations can be determined from data.

3. From the energy gradient to one MD step

$$\mathbf{F}_i(\mathbf{R}) = -\nabla_i U(\mathbf{R}), \qquad m_i \frac{d^2\mathbf{R}_i}{dt^2} = \mathbf{F}_i(\mathbf{R})$$

The negative coordinate derivative of the energy gives the force on each atom. Energy is a scalar associated with one configuration; forces are vectors, with a magnitude and direction for every atom. Taking the direction of increasing separation r at atom i as positive, the radial force for an LJ pair is

$$f_r(r) = -\frac{dU^{\mathrm{LJ}}}{dr} = \frac{24\epsilon}{r}\left[2\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

For r<2^(1/6)σ, the force is repulsive and separates the atoms; at larger separations, it is attractive. At σ, the energy is zero but the force is 24ε/σ, which is nonzero. The force depends on the energy gradient rather than its absolute value. The enlarged panel on the right of Fig. 1 shows the otherwise small-looking attractive region.

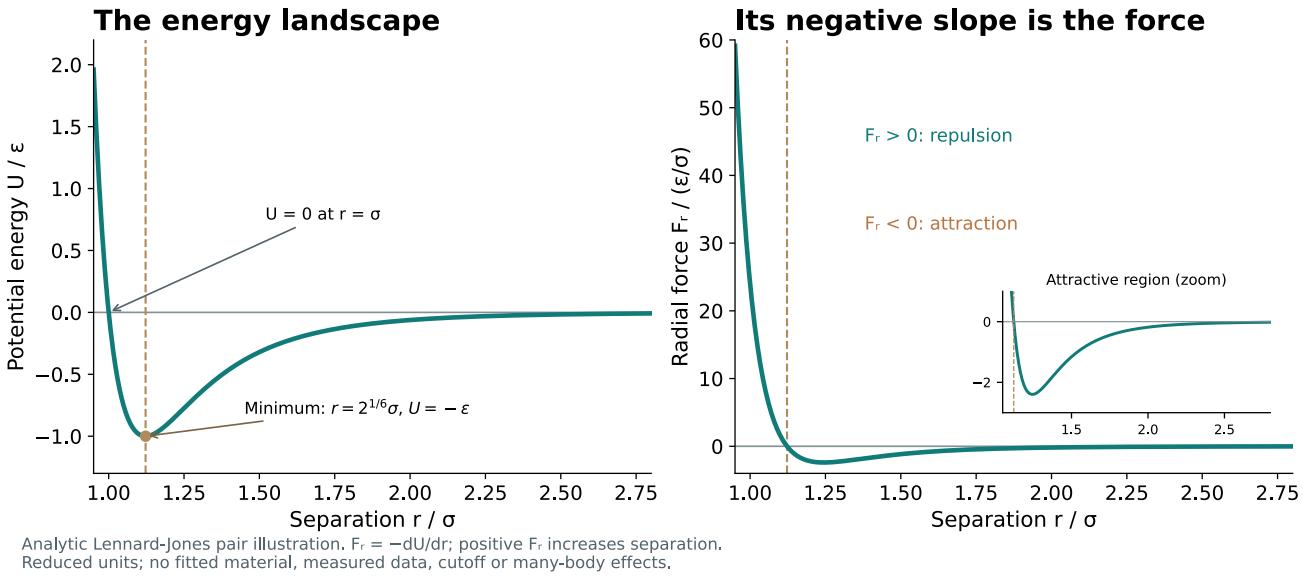


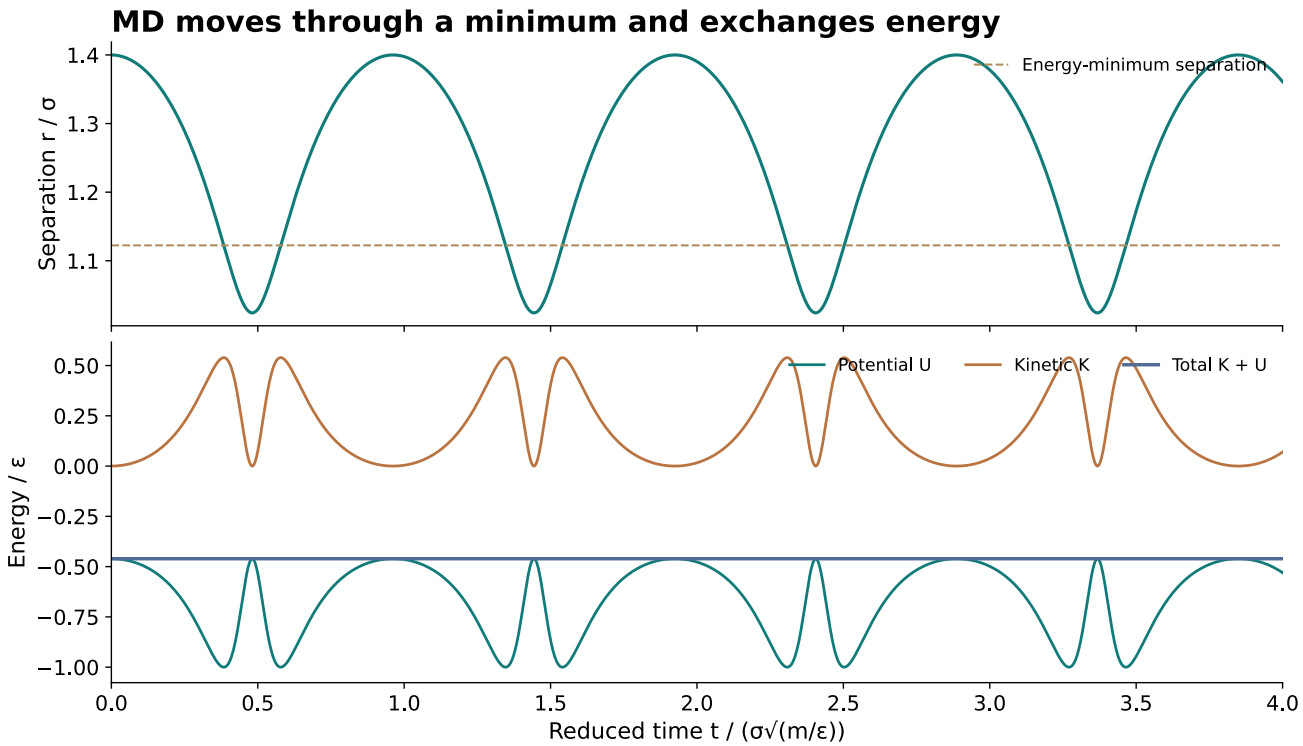
Figure 1. Energy and negative distance derivative of the same LJ function. Here r is the interatomic separation, and positive f_r denotes repulsion in the direction of increasing separation. Quantities are normalized by ε and σ; the small panel on the right enlarges the attractive region. Analytical calculation for this review, checked by numerical differentiation of the energy. The function was not fitted to a real material. Equation source: [3].

In code, one calculates the displacement $d_{ij}=R_i-R_j$ and distance $r=|d_{ij}|$, adds $f_r d_{ij}/r$ to the force on atom i, and adds the opposite force to atom j. For a many-body potential, differentiation must include every coordinate dependence of the energy on the surrounding atomic environments. Automatic differentiation of an ML energy model performs this calculation as well. The force on an atom generally cannot be obtained by differentiating only the energy contribution assigned to that atom.

An MD integrator advances positions and velocities using these forces and the atomic masses. One velocity Verlet step advances the velocities by half a step, updates the positions with those velocities, recalculates forces for the new complete configuration, and advances the velocities through the remaining half step. [6]

$$\mathbf{v}_i(t+\Delta t/2) = \mathbf{v}_i(t) + \frac{\Delta t}{2m_i}\mathbf{F}_i(t)$$
$$\mathbf{R}_i(t+\Delta t) = \mathbf{R}_i(t) + \Delta t\,\mathbf{v}_i(t+\Delta t/2)$$
$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t+\Delta t/2) + \frac{\Delta t}{2m_i}\mathbf{F}_i(t+\Delta t)$$

In Fig. 2, two equal-mass LJ particles start at rest, farther apart than their equilibrium separation. Attraction accelerates them toward one another. They pass the energy minimum, move closer, and then separate again under repulsion. Kinetic and potential energy exchange while their sum remains constant within a small integration error. MD follows time-dependent motion with velocity and inertia, unlike energy minimization.



Review calculation: isolated equal-mass LJ pair, r(0)=1.4σ, relative velocity 0, velocity Verlet, Δt=0.001.
No thermostat, damping, material fit or many-body environment. Finite-step energy error was checked by halving Δt.

Figure 2. Educational one-dimensional MD calculation for an isolated equal-mass LJ pair. The initial separation is 1.4σ and the initial relative velocity is zero. Each particle has mass m. Velocity Verlet is used with a reduced time unit σ√(m/ε) and a time step of 0.001. There is no thermostat, damping, or periodic boundary condition. Convergence of the total-energy error was checked by halving the time step. This is not a reproduction of a real molecule or an SO3LR result.

Large systems reduce the cost of searching all atom pairs by updating neighbor lists with a cutoff and an additional buffer. [7] Long-range electrostatics under periodic boundary conditions are combined with a separate summation method such as PME. [8] If a switching function multiplies the energy, its derivative must also enter the force. The time step, unit system, boundary conditions, and choice of thermostat and barostat remain implementation requirements when using an MLFF. An NVT system exchanges energy with a heat bath, so the isolated-system test for conservation of K+U does not apply unchanged.

4. Learning the energy function to be evaluated

Electronic-structure calculations determine the energy and forces for a selected electronic state at a given atomic configuration. Actual DFT results depend on the exchange-correlation functional, basis and pseudopotentials, charge and spin states, treatment of dispersion and relativity, and convergence settings. [9] Repeating this calculation at every MD step in Born-Oppenheimer dynamics can describe flexible chemistry, but is expensive. Simulating 1 μs with a 1 fs time step requires a billion force evaluations.

An MLFF fits a surrogate function to energies and forces computed in advance for representative structures. A typical local energy model converts the environment of atom i into a descriptor D_i and sums atomic contributions $e\theta(D_i)$. [10] Here θ denotes learned coefficients or neural-network weights. Atomic contributions

are an internal decomposition used to represent the total energy; they do not imply that the energy of each atom is separately measured in an experiment and supplied as a target.

$$U_\theta(\mathbf{R}) = \sum_i e_\theta\big(D_i(\mathbf{R})\big) + U_{\text{explicit}}(\mathbf{R})$$
$$\mathbf{F}_{\theta,i}(\mathbf{R}) = -\nabla_i U_\theta(\mathbf{R})$$

Uexplicit contains selected explicit contributions such as long-range electrostatics, dispersion, or short-range repulsion. Models need not use this decomposition or include all these terms. The energy must remain unchanged if the configuration is translated or rotated, or if labels of atoms of the same element are permuted. Force vectors must rotate with the configuration. Building these symmetries into a model removes irrelevant differences that would otherwise have to be learned. Equivariant models retain directional internal information, providing a way to learn angular dependence while satisfying these conditions. [11]

Joint training on energy and force labels can be described by an objective such as the one below. MSE is the mean squared error over the selected structures and components, and W denotes the stress or virial as defined in the implementation. This illustrative expression captures a concept shared by several approaches.

$$\begin{aligned}\mathcal{L}(\theta) = {} & w_E\ \mathrm{MSE}(U_\theta, U_{\text{ref}}) \\ & + w_F\ \mathrm{MSE}(-\nabla_{\mathbf{R}} U_\theta, \mathbf{F}_{\text{ref}}) \\ & + w_W\ \mathrm{MSE}(W_\theta, W_{\text{ref}})\end{aligned}$$

Training requires decisions about whether to divide energies by the number of atoms, average forces over 3N components, and exclude missing labels. The values and units of wE, wF, and wW, together with weights assigned to different data sources, change the compromise the model learns. Updating θ reduces this objective. For a new configuration, the model evaluates Uθ and its derivatives without repeating the electronic-structure calculations used in training. Expressive representations, repeated reuse, and transfer learning are practical strengths of MLFFs. Accuracy for configurations or physics absent from the data requires separate validation.

Conventional ML-based MD and Born-Oppenheimer MD propagate nuclear motion classically. Learning an energy surface that reflects electronic quantum mechanics does not also describe nuclear zero-point motion or tunneling. Problems where nuclear quantum effects matter require additional treatment, such as path integrals. Equilibrium nuclear quantum statistics and real-time quantum dynamics also involve different approximations. [12,13]

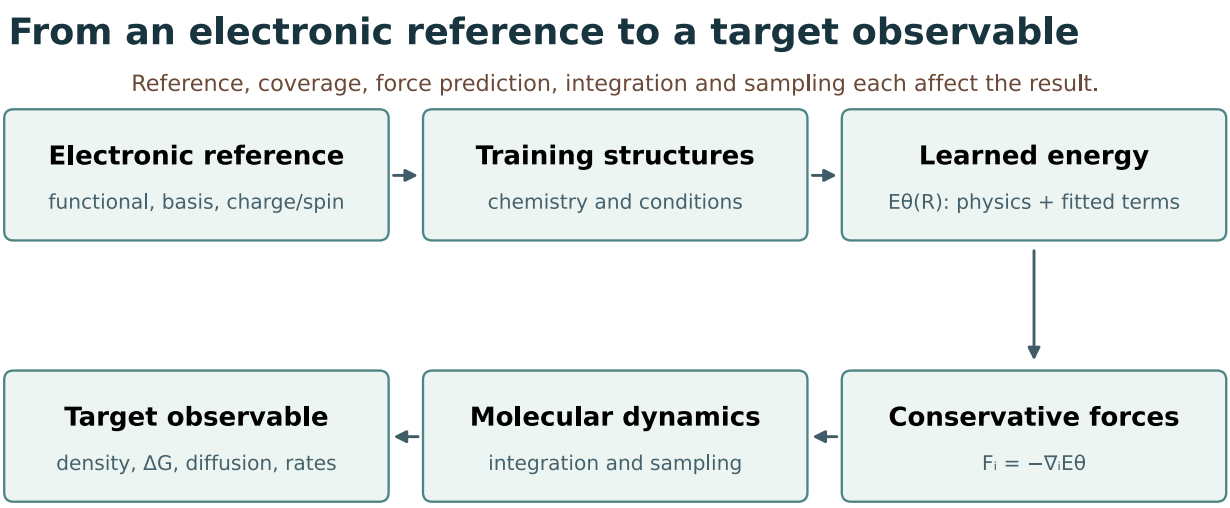


Figure 3. From the electronic-structure reference and training configurations to an energy function, forces, MD sampling, and the target property. Approximations in the electronic-structure reference, learning error, integration, and sampling are checked at different stages. Explanatory schematic created for this review. Sources for the relationships: [9,12,13].

5. Models differ in representation and training scope

Choosing an MLFF begins with examining what information represents a configuration and how that information produces an energy. Even within one model family, changes to the data, cutoff, and model size alter accuracy and cost. The following table compares representative design principles; it is not a performance ranking.

Design and representative models	What is learned	Strengths and conditions for use
Kernels: GAP and sGDML	GAP performs regression using similarities between local atomic environments. sGDML learns the full force field of a specific molecule while enforcing energy conservation. [14,12]	Physical symmetries and the regression structure can make efficient use of expensive reference data. The number of representative environments and the training domain determine cost and transferability; the two models use different spatial decompositions.
Symmetry descriptors and neural networks: Behler-Parrinello and ANI	Atomic neural-network weights are fitted using invariant distance- and angle-based descriptors, and the atomic contributions are summed. [10,15]	A fixed network evaluates new configurations. Its representational scope depends on descriptors, cutoff, elements, and training environments. The H/C/N/O coverage of the original ANI-1 should not be generalized to every ANI version.
Many-body basis expansions: ACE	Atomic environments are expanded in a symmetry-adapted basis. A linear energy model with a fixed basis can be fitted by regression of its coefficients. [16]	Expansion order and basis size tune expressivity and cost. ACE is not restricted to linear models; nonlinear constructions are also possible.

Design and representative models	What is learned	Strengths and conditions for use
Equivariant graph neural networks: NequIP and MACE	Atoms and neighbors form a graph whose features, including directional information, are updated. MACE constructs higher-order many-body messages. [11,17]	These models learn complex angular and environmental dependence. Channels, angular order, layers, and neighbor counts affect memory and cost. A finite receptive range and correct long-range asymptotic behavior are separate considerations.

Training strategy and the choice of physical terms provide additional distinctions that overlap with this table. Pretrained potentials such as CHGNet and MACE-MP use weights learned from broad datasets either directly or as starting points for fine-tuning on a new system. [18,19] Pretraining does not itself denote a new force law or a distinct neural-network architecture. Using a model for inference, refitting its weights to new labels, and training from scratch are different operations.

PhysNet and SO3LR Tauri illustrate approaches that combine learned energies and environment-dependent properties with explicit electrostatics, dispersion, and other terms. [20,21] Long-range terms allow selected interactions beyond the graph cutoff to be calculated, but their functional forms and assumptions about charge and polarization still matter. Pretraining, equivariance, and explicit long-range terms can coexist in one model.

Opening up the Tauri calculation makes the role of learning more concrete. The network predicts atomic semi-local energies, charges, and polarizability and dispersion-coefficient ratios. These outputs enter explicit terms in the total energy. Holding the elements Z, total charge Q, spin multiplicity S, and trained parameters θ fixed, we can write [21,31]

$$
\begin{aligned}
E_\theta(\mathbf{R}) = {} & E^{\mathrm{NLH}}_{\mathrm{rep}}(\mathbf{R}) + E_{\mathrm{SO3k},\theta}(\mathbf{R}) \\
& + E_{\mathrm{elec}}[\mathbf{R}, q_\theta(\mathbf{R})] \\
& + E_{\mathrm{disp}}[\mathbf{R}, \alpha_\theta(\mathbf{R}), C_{6,\theta}(\mathbf{R})]
\end{aligned}
$$

The NLH repulsion uses pair-specific tables fitted to separate DFT calculations and held fixed during ML training. The semi-local energy is fitted with the other terms to the total reference energy. This is different from first fitting a complete energy and then adding Coulomb and dispersion energies indiscriminately: the learned and explicit contributions must be defined together to avoid double counting. [21]

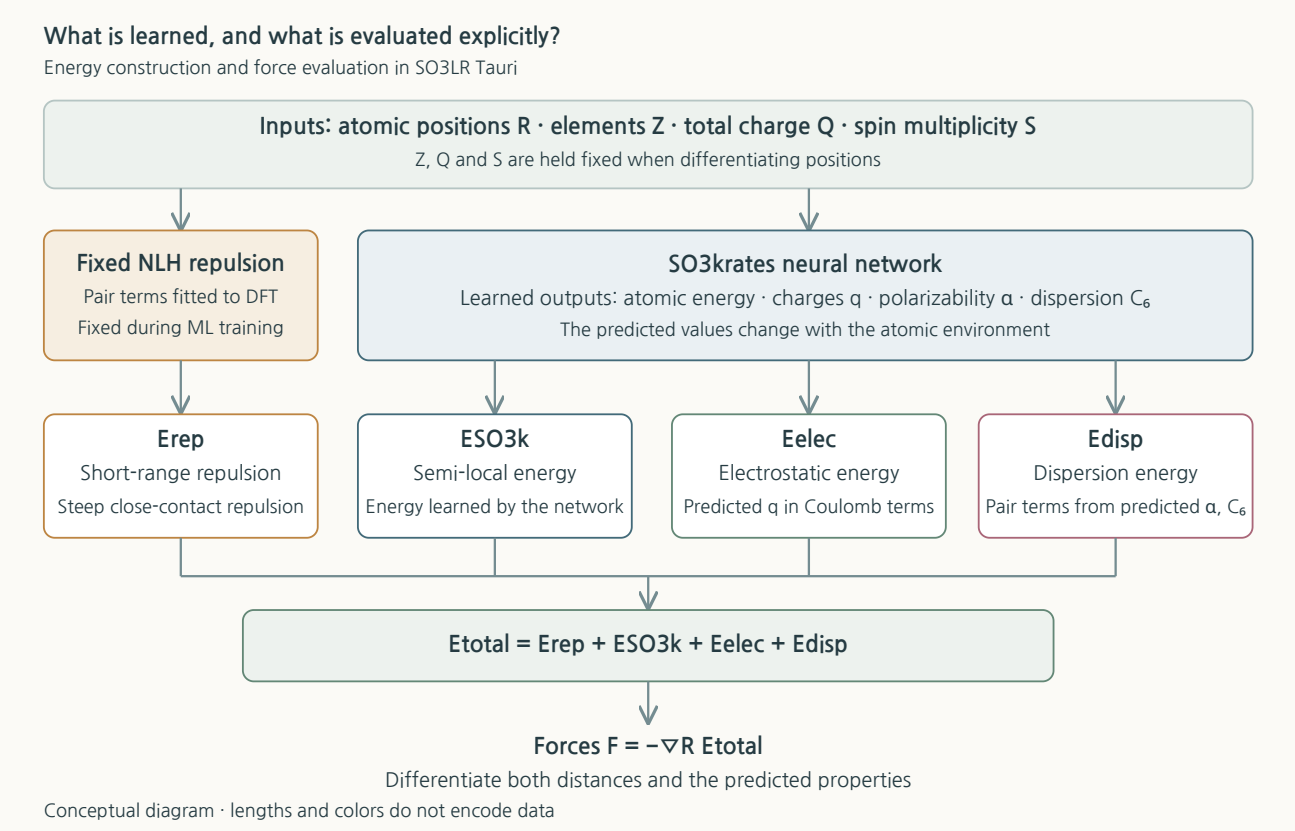


Figure 4. Fixed NLH repulsion and network outputs combine into four Tauri energy terms. Charges, polarizabilities, and dispersion coefficients depend on coordinates, so their responses also enter the forces. Original conceptual diagram based on paper Sections II.1/S1 and a fixed code version; it is not quantitative performance data. [21,31]

Force evaluation differentiates both interatomic distances and the predicted charges. For the electrostatic contribution alone, the chain rule gives the following expression. The first term holds charges fixed; the second accounts for changes in the charge predictions as the geometry changes. [31]

$$\nabla_i E_{\mathrm{elec}} = \nabla_i E_{\mathrm{elec}}\big|_q + \sum_j \frac{\partial E_{\mathrm{elec}}}{\partial q_j} \nabla_i q_{\theta,j}$$

The negative of this derivative is the electrostatic force contribution. The same principle applies to polarizabilities and dispersion coefficients, with the cell and boundary conditions also held fixed. The basic implementation differentiates the complete model energy with respect to positions; the fast path also backpropagates the responses of predicted properties. This statement comes from static inspection of the public code, not a new MD run. [31]

Type of setting	Examples	What it determines
Learned parameters	Kernel and basis coefficients; neural-network weights and element embeddings; weights of output layers predicting charge and dispersion properties	The fitted function θ within the selected representation and data
Representation and training settings	Cutoff; basis, many-body, and angular orders; number of layers and channels; regularization; objective and data weights	Representational scope, training compromises, and computational cost. Depending on the implementation, some basis parameters may also be learned.
Physical reference for the labels	Electronic-structure method, basis, charge and spin, dispersion corrections, and convergence conditions	What the model learns as its target. Changing the reference changes the problem even for the same structure.

A model name alone does not specify these three sets of choices. The architecture name MACE should be distinguished from the training data and checkpoints of MACE-MP, and the selected version, reference level, and units should be recorded. With limited data for an application, a relevant pretrained model can be validated and fine-tuned. Unsupported chemistry or a different electronic-structure reference may instead motivate training a model from scratch for a narrower system. These options should be compared against the same objective and validation data.

6. Training data define the chemistry and conditions

Expressivity alone leaves a problem if the configurations that must be learned are absent from the data. Coverage must be examined for bonding and coordination, charge and spin, compression and thermal deformation, solvents and ions, and reaction pathways, as well as for elements. The labels actually used in training—energy, forces, or stress—and the accuracy of the reference calculations also need to be checked. SO3LR Tauri, a preprint released on 4 November 2026, provides a concrete example of how these choices are made.

Data release 1.0.0 lists 147,894,100 training, 499,644 validation, and 499,672 test entries: 148,893,416 in total. The paper's roughly 148 million training samples and the roughly 148.9 million all-split total have different denominators. The neutral-water pool reuses 694,467 structures from other sources, so the aggregate is not a count of unique geometries. Subtracting that known reuse would still not establish complete deduplication across all sources. [22]

Dataset	Chemistry and labels included	Conditions for interpretation
QCML	Small molecules and distorted structures. PBE0+MBD-NL energies, forces, and electronic properties	Broad elemental coverage does not mean uniform coverage of element combinations, conformations of large molecules, or diverse high-spin states. [23]
QCell	Biomolecular fragments including lipids, sugars, nucleic acids, and ions. PBE0+MBD(-NL)	Reinforces recurring local environments in large biomolecules. Collecting fragments does not validate the equilibrium distribution of an entire protein. [24]
OMol25-0 and OPoly26	Diverse charged, metal-containing, and reactive environments, and polymer fragments. ωB97M-V force labels	Tauri does not mix energy labels from these two sources into training. Elemental, charge, and distorted-structure coverage, as well as the version, must be checked separately. [22,25,26]

The levels of theory must also be read together with the training procedure. Initially, a shared representation feeds two reference-specific output heads. Later, a production head initialized from the PBE0 branch and the shared representation are updated with both levels. OMol25-0, OPoly26, and the reused water pool contribute force targets only; unavailable labels are masked out of the loss. Direct atomic polarizability and dispersion-related targets come from designated PBE0+MBD-NL data. The final model is therefore best understood through its targets, stages, and weights, rather than as a simple copy of one DFT reference. [21,22]

Dataset size and exposure during training are different quantities. A small source can be sampled frequently, while a large set of similar conformers can be downweighted. Tauri's inclusion of 85 elements does not make that distribution uniform. Applications still require checking the combinations of elements, charges, spins, coordination environments, and condensed-phase configurations represented in training and validation. [21,22]

Using the same structures does not imply using the same labels. For example, the main electronic-structure reference in the original SPICE dataset is ωB97M-D3(BJ), whereas the SPICE dipeptide structures included in Tauri were recalculated at PBE0+MBD. [22,27] Inferring the training reference from a dataset name alone misses such differences.

Test splits must also be interpreted in relation to their purpose. Some internal Tauri data are split by randomly shuffling structures, whereas the reported OMol and OPoly force evaluations use the original datasets' validation splits based on atomic composition. [22,25,26] Transfer to molecules of different composition, transfer to new temperatures, phases, or reaction pathways, and stability over long trajectories require different tests.

Biomolecular and crystalline-material datasets are not interchangeable general-purpose resources. SPICE and Nutmeg primarily target room-temperature configurations of drug-like molecules and peptides, and Nutmeg does not target chemical reactions. [27] The MPtrj data used by CHGNet are based on DFT relaxation

trajectories of inorganic crystals, with splits by material ID. [18] A model that performs well for molecules in aqueous solution need not be the best choice for crystal phase stability or defects.

7. Applications depend on the material and the question

In materials science and chemistry, force fields are used to calculate optimized structures, vibrations and elasticity, defect migration, and thermal and ionic transport. Molecular chemistry examines configurations, spectra, and reaction pathways; liquids and electrolytes raise questions about solvation, ion pairing, and diffusion; biomolecular studies consider structural distributions, protonation, and binding. Each question requires particular interactions and sampling.

Fast local potentials estimate energy from neighboring atomic environments. They are efficient at a specified density and cutoff, but appropriate long-range treatment is needed when distant charges or collective electronic responses matter. Neural-network potentials that explicitly address nonlocal charge transfer provide one example of resolving this issue through model design. [28]

For example, keep the local inputs determining an atom's force unchanged while rearranging ions beyond their finite receptive field. The electric field on that atom can change, but a prediction depending only on those inputs has no information about the changed external field. Capturing that difference requires suitable long-range terms or environmental response. Adding more examples and changing the information available to the model address different limitations.

Problem	Required physics and properties to check	Implications for model choice
Covalent solids within a specified phase; screened metals	Bonding, deformation, defects, and vibrations. Revalidate for surfaces or other phases.	A local ML potential validated within that domain can be an efficient candidate.
Electrolytes, proteins, and charged interfaces	Long-range electrostatics, dielectric response, ion pairing, and solvation	Examine explicit long-range terms and charge and environmental responses. Including PME alone does not fully describe polarization.
Molecular crystals, adsorption, and molecular recognition	Competition among dispersion, hydrogen bonding, solvent interactions, and intramolecular interactions	Check the dispersion functional form, condensed-phase data, and relative binding and solvation free energies.
Bond formation, bond breaking, and proton transfer	Charge and spin states, transition states, and competing pathways	Data covering reactive environments and barrier validation are needed. The absence of a fixed bond topology alone does not guarantee accurate reaction rates.

Tauri corrects predicted charges by distributing the residual charge equally across atoms so their sum matches the input Q. This global algebraic correction differs from charge equilibration through energy minimization. Predicted polarizabilities and dispersion coefficients construct environment-dependent pair dispersion terms. MBD in a training reference does not mean that inference separately solves self-consistent induced dipoles or long-range many-body dispersion. An overall spin input is also distinct from spin dynamics. [21,31] Excited-state photochemistry and nonadiabatic transitions require checking whether a scalar potential for one electronic state is sufficient.

Empirical force fields have also been carefully tuned to quantum-chemical and experimental data, and include polarizable and reactive models. When long-time sampling is important for a validated biomolecular problem with fixed bond topology, a well-validated empirical model remains an important comparator. Comparing the total cost of reaching a target error gives no inherent priority to the MLFF category itself.

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8. Working backward from the property to the required accuracy

The roughly 1 kcal/mol commonly called chemical accuracy is a useful scale rooted in thermochemistry. It is not a universal certification threshold for forces, vibrational frequencies, reaction rates, or diffusion coefficients. [12] For an entire reaction or pair of states being compared, this energy corresponds to about 43.36 meV. An overall test error normalized in meV/atom cannot be compared directly with the error in the energy difference between a reactant and a transition state.

When the target is a reaction rate, the tolerance can be specified more concretely. Assuming the same prefactor and transmission coefficient in transition-state theory, an error $\delta\Delta G^{\ddagger}$ in the free-energy barrier changes the rate as follows:

$$\frac{k_{\mathrm{ML}}}{k_{\mathrm{ref}}} = \exp\left[-\frac{\delta\Delta G^{\ddagger}}{RT}\right]$$

Here R is the gas constant and T the absolute temperature. At 300 K, RT is about 0.596 kcal/mol, so overestimating or underestimating a free-energy barrier by 1 kcal/mol changes the rate by a factor of about 5.35. Keeping the effect of barrier error alone within a factor of two requires an error no larger than about 0.413 kcal/mol. This is an illustrative analytical calculation for this review, not a conversion of mean energy MAE into an actual rate error.

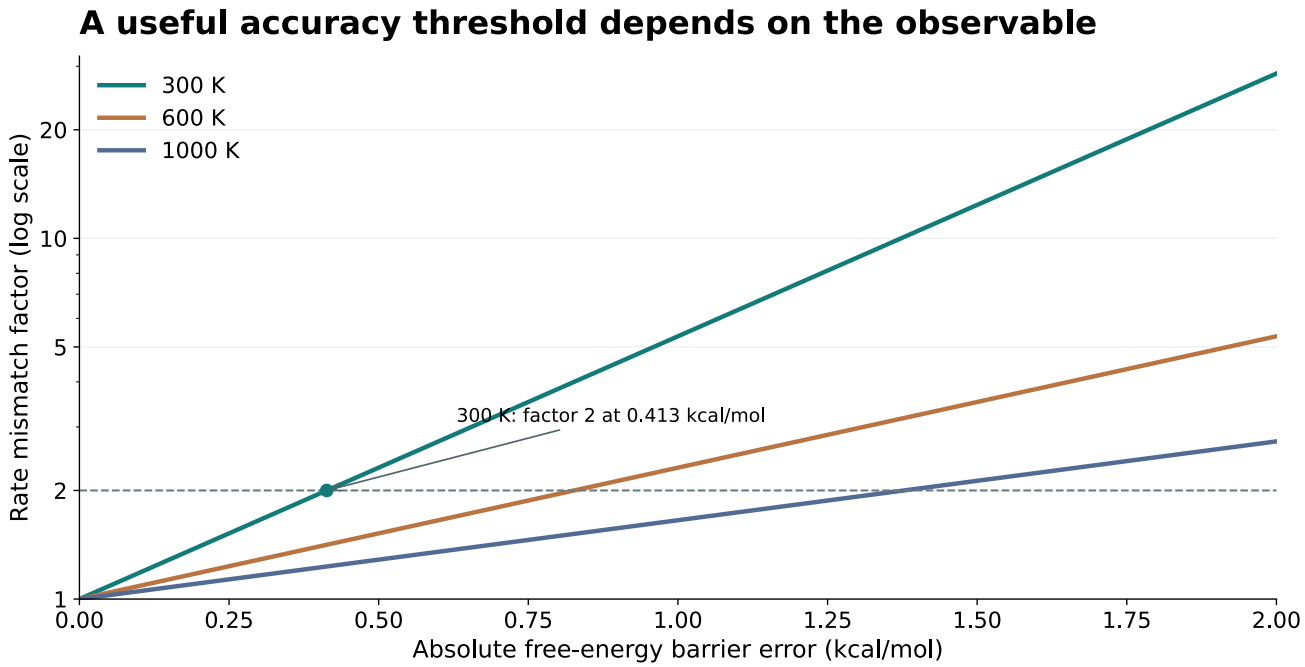


Figure 5. Rate-change factor exp(|δΔG‡|/RT) produced by free-energy barrier error, assuming the same prefactor. The vertical axis is logarithmic, and curves compare 300, 600, and 1000 K. This educational calculation for the review shows how the meaning of 1 kcal/mol depends on temperature and the observable. It is not an MLFF benchmark or a measured barrier.

For a liquid-density target, equilibrium density and its temperature and pressure dependence matter. Ionic conductivity requires diffusion, ion pairing, and correlations in collective transport; a crystalline phase transition requires free-energy differences between competing phases. Biomolecular binding or folding requires solvation, relative free energies of metastable states, and sufficient sampling. Once the target property and conditions are specified, one can work backward to identify which errors in energy, forces, stress, or reaction barriers must be reduced. This establishes a minimum acceptable accuracy and tolerance for the particular problem.

This minimum accuracy is not a fixed ranking of model families. The same architecture can pass or fail depending on its reference data, checkpoint, charge state, and observable. If the representation omits relevant long-range information, or the electronic reference itself does not resolve the target barrier accurately enough, improving the fit of the same function does not remove that cause. The error tolerance must be considered together with what limits the calculation.

9. What remains to validate after fitting the forces

The Forces are not Enough benchmark shows why low test force error, stable trajectories, and accurate properties must be checked separately. On Water-10k, NequIP produced longer stable trajectories than GemNet-T, even though GemNet-T had a lower force MAE. Other models that remained stable still failed to reproduce diffusion accurately. The water reference here is the flexible SPC/E model rather than DFT. These results therefore demonstrate disagreement among evaluation metrics; they do not establish a universal ranking of DFT-based models. [29]

The Tauri examples also need to be read by observable and reference. The following two reported numbers answer different physical questions. [21]

Reported quantity	Value and reference	Application-specific check
Liquid-water diffusion	T-M at300 K, Yeh-Hummer-corrected119 Å²/ns; experiment at≈298 K:230 Å²/ns	Validate transport separately from liquid structure or density.
T-L proline-catalyzed aldol pathway	Electronic activation-barrier MAE 2.9 kcal/mol against PBE0+MBD-NL single-point energies at T-L geometries	Do not substitute this electronic-barrier error for a free-energy barrier error to infer an actual rate error.

Finding reaction pathways and maintaining structures are valuable advances. Predicting properties additionally requires relative pathway energies, solvent and entropic contributions, and sufficient sampling. The case is useful because it exposes both a design that broadens accessible calculations and the errors that remain for target observables.

A systematic application can use validation designed around its purpose. The procedure below is a proposed computational approach drawn from prior studies and this case, rather than an independent reproduction of the source paper's experiments.

1. Specify the target property and its tolerance. Record the elements, charge and spin, phase, concentration, temperature, pressure, boundary conditions, and units of the property.
2. Select representative and challenging structures separately, and begin by checking the electronic-structure reference. For reactions, include transition states and competing pathways as well as reactants; for condensed phases, include ionic, solvent, compressed, and distorted environments.
3. Separate training from validation and examine both average errors and the high-error tail. Check energy differences, forces, and any required stresses and barriers under the intended configurational conditions.
4. Use short NVE calculations to check drift as a function of time step, cutoff, and long-range accuracy, and determine whether a thermostat is masking integration problems. Energy conservation and an accurate potential are separate requirements.
5. Estimate uncertainty in the target property using independent trajectories and appropriate correlation times. Check convergence with box size, trajectory length, free-energy windows, and treatment of nuclear motion as required for the problem.
6. Identify and address distinct causes of failure. Missing environments call for targeted data and retraining; missing physics requires an appropriate model; slow transitions require a sampling strategy. After revision, validate both the original and new domains again.

Active learning can reduce the cost of adding reference calculations for new environments. FLARE provides a concrete precedent: it calls electronic-structure calculations to update the model when uncertainty grows. [30] Uncertainty is a signal to request a reference calculation, however, not a guarantee of accuracy. Several models

can share errors outside the data distribution, so agreement among them alone cannot establish accuracy.

An application record should include the fixed checkpoint, code commit and file hash, units, total charge and spin, cutoffs, PME, precision, and timestep. A short model alias in the official repository resolves to a particular checkpoint and can change later. Also distinguish the MIT code terms from the Tauri weights' CC BY-NC 4.0 terms when checking suitability for the intended use. [31]

10. Accounting for atom count and time scale together

Computational cost includes reference-data generation and training, force evaluation at every MD step, the integration interval, replicate trajectories, and free-energy sampling. When reusing a pretrained model, one can budget separately for training costs already incurred and the validation, fine-tuning, and production costs of the new problem. Local models often scale linearly with the number of atoms at fixed density and cutoff, whereas FFTs for long-range PME generally carry an N log N cost. [8]

> Cost for a new problem ≈ reference calculations + validation and fine-tuning + (physical time/integration interval) × cost per step + replicates and analysis

Tauri's medium model is reported to take approximately 0.98 μs/atom/step on an MI355X with 288 GB of memory. This figure excludes JIT compilation and uses periodic water, PME, and the fast memory mode. For 50,000 atoms and a 1 fs time step, it corresponds to about 1.8 ns/day. The maximum capacity of 84 million atoms is a different measure from throughput for long biomolecular trajectories. [21]

For example, integrating 1 μs with a 1 fs time step requires a billion steps. A simple extrapolation at the stated converted throughput gives about 556 GPU-days for 1 μs. This is an arithmetic illustration assuming that the water-benchmark throughput is sustained, not a runtime prediction for an actual protein. Independent replicates, multiple enhanced-sampling windows, or multiple path-integral beads would change the required computation.

Speed comparisons must match the GPU and memory, numbers of atoms and neighbors, precision, cutoff and PME settings, time step, and inclusion of JIT compilation and initialization. If a faster model increases bias in the target free energy or requires longer trajectories, a lower cost per step may not reduce the cost of the final answer. The relevant comparison is the total cost of obtaining the desired property within the required uncertainty.

11. From atomic forces to the answer sought

Choosing an energy function, obtaining forces from its gradient, and using those forces to advance atomic motion form a common structure linking empirical force fields and many MLFFs. MLFFs change how complex energy functions are constructed and reused. Environmental dependence learned from electronic-structure

calculations and flexible representations create opportunities to explore more configurations and materials.

Turning that opportunity into a computable answer requires a precise question. Finding a reaction pathway and reproducing its rate demand different data, accuracy, and time scales, as do briefly maintaining a protein structure and determining its folding equilibrium. Specify the property of interest, choose a model capable of representing the relevant physics, and validate it with independent data and converged sampling. This order makes it possible to compare force-field types and sizes, data-generation costs, and computational budgets against the same scientific objective.

References

[1] A. Rahman, Correlations in the motion of atoms in liquid argon, Phys. Rev. 136, A405 (1964). doi:10.1103/PhysRev.136.A405

[2] OpenMM contributors, Standard forces, OpenMM User Guide 8.6 (accessed 6 November 2026). https://docs.openmm.org/latest/userguide/theory/02_standard_forces.html

[3] LAMMPS developers, pair_style lj/cut command, LAMMPS release 30Sep2026 (accessed 6 November 2026). https://docs.lammps.org/pair_lj.html

[4] M. S. Daw and M. I. Baskes, Embedded-atom method: Derivation and application to impurities, surfaces, and other defects in metals, Phys. Rev. B 29, 6443 (1984). doi:10.1103/PhysRevB.29.6443

[5] J. Tersoff, New empirical approach for the structure and energy of covalent systems, Phys. Rev. B 37, 6991 (1988). doi:10.1103/PhysRevB.37.6991

[6] LAMMPS developers, How a timestep works, LAMMPS release 30Sep2026 (accessed 6 November 2026). https://docs.lammps.org/Developer_flow.html

[7] LAMMPS developers, Neighbor lists, LAMMPS release 30Sep2026 (accessed 6 November 2026). https://docs.lammps.org/Developer_par_neigh.html

[8] T. Darden, D. York, and L. Pedersen, Particle mesh Ewald: An N log(N) method for Ewald sums in large systems, J. Chem. Phys. 98, 10089 (1993). doi:10.1063/1.464397

[9] W. Kohn and L. J. Sham, Self-consistent equations including exchange and correlation effects, Phys. Rev. 140, A1133 (1965). doi:10.1103/PhysRev.140.A1133

[10] J. Behler and M. Parrinello, Generalized neural-network representation of high-dimensional potential-energy surfaces, Phys. Rev. Lett. 98, 146401 (2007). doi:10.1103/PhysRevLett.98.146401

[11] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, and B. Kozinsky, E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials, Nat. Commun. 13, 2453 (2022). doi:10.1038/s41467-022-29939-5

[12] S. Chmiela, H. E. Sauceda, K.-R. Müller, and A. Tkatchenko, Towards exact molecular dynamics simulations with machine-learned force fields, Nat. Commun. 9, 3887 (2018). doi:10.1038/s41467-018-06169-2

[13] i-PI developers, Core features, i-PI documentation (accessed 6 November 2026). https://docs.ipi-code.org/features.html

[14] A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi, Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons, Phys. Rev. Lett. 104, 136403 (2010). doi:10.1103/PhysRevLett.104.136403

[15] J. S. Smith, O. Isayev, and A. E. Roitberg, ANI-1: An extensible neural network potential with DFT accuracy at force field computational cost, Chem. Sci. 8, 3192 (2017). doi:10.1039/C6SC05720A

[16] R. Drautz, Atomic cluster expansion for accurate and transferable interatomic potentials, Phys. Rev. B 99, 014104 (2019). doi:10.1103/PhysRevB.99.014104

[17] I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csanyi, MACE: Higher order equivariant message passing neural networks for fast and accurate force fields, Adv. Neural Inf. Process. Syst. 35, 11423 (2022). https://papers.neurips.cc/paper_files/paper/2022/hash/4a36c3c51af11ed9f34615b81edb5bbc-Abstract-Conference.html

[18] B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, and G. Ceder, CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling, Nat. Mach. Intell. 5, 1031 (2023). doi:10.1038/s42256-023-00716-3

[19] I. Batatia et al., A foundation model for atomistic materials chemistry, J. Chem. Phys. 163, 184110 (2025). doi:10.1063/5.0297006

[20] O. T. Unke and M. Meuwly, PhysNet: A neural network for predicting energies, forces, dipole moments, and partial charges, J. Chem. Theory Comput. 15, 3678 (2019). doi:10.1021/acs.jctc.9b00181

[21] A. Kabylda, F. N. Brünig, M. Gallegos, J. T. Frank, E. Moerman, O. T. Unke, S. Chmiela, K.-R. Müller, and A. Tkatchenko, SO3LR Tauri: Generally applicable force fields for reactive (bio)molecular simulations covering the periodic table, arXiv:2611.06237v1 (2026). https://arxiv.org/abs/2611.06237v1

[22] A. Kabylda, F. Brünig, M. Gallegos González, T. Frank, E. Moerman, O. T. Unke, S. Chmiela, K.-R. Mueller, and A. Tkatchenko, Data for "SO3LR Tauri: Generally Applicable Force Fields for Reactive (Bio)Molecular Simulations Covering the Periodic Table", version 1.0.0, Zenodo (2026). doi:10.5281/zenodo.22920714

[23] S. Ganscha, O. T. Unke, D. Ahlin, H. Maennel, S. Kashubin, and K.-R. Müller, The QCML dataset, quantum chemistry reference data from 33.5M DFT and 14.7B semi-empirical calculations, Sci. Data 12, 406 (2025). doi:10.1038/s41597-025-04720-7

[24] A. Kabylda, S. Suárez-Dou, N. Davoine, F. N. Brünig, and A. Tkatchenko, QCell: Comprehensive quantum-mechanical dataset spanning diverse biomolecular fragments, AI Sci. 2, 025003 (2026). doi:10.1088/3050-287X/ae5267

[25] D. S. Levine et al., The Open Molecules 2025 (OMol25) dataset, evaluations, and models, arXiv:2505.08762v2 (2026). https://arxiv.org/abs/2505.08762v2

[26] D. S. Levine, N. Liesen, L. Chua, J. Diffenderfer, H. I. Ingolfsson, M. P. Kroonblawd, N. Kumar, A. Maiti, S. S. Mohottalalage, M. Shuaibi, B. Van Essen, B. M. Wood, C. L. Zitnick, S. M. Blau, and E. R. Antoniuk, The Open Polymers 2026 (OPoly26) dataset and evaluations, arXiv:2512.23117v2 (2026). https://arxiv.org/abs/2512.23117v2

[27] P. Eastman, B. P. Pritchard, J. D. Chodera, and T. E. Markland, Nutmeg and SPICE: Models and data for biomolecular machine learning, J. Chem. Theory Comput. 20, 8583 (2024). doi:10.1021/acs.jctc.4c00794

[28] T. W. Ko, J. A. Finkler, S. Goedecker, and J. Behler, A fourth-generation high-dimensional neural network potential with accurate electrostatics including non-local charge transfer, Nat. Commun. 12, 398 (2021). doi:10.1038/s41467-020-20427-2

[29] X. Fu, Z. Wu, W. Wang, T. Xie, S. Keten, R. Gomez-Bombarelli, and T. Jaakkola, Forces are not enough: Benchmark and critical evaluation for machine learning force fields with molecular simulations, Trans. Mach. Learn. Res. (2023). https://openreview.net/forum?id=A8pqQipwkt

[30] J. Vandermause, S. B. Torrisi, S. Batzner, Y. Xie, L. Sun, A. M. Kolpak, and B. Kozinsky, On-the-fly active learning of interpretable Bayesian force fields for atomistic rare events, npj Comput. Mater. 6, 20 (2020). doi:10.1038/s41524-020-0283-z

[31] General Molecular Simulations, SO3LR, GitHub, commit ee62fa28860850d736845c3b91fba051d230f478 (2026). https://github.com/general-molecular-simulations/so3lr/tree/ee62fa28860850d736845c3b91fba051d230f478

How this review was made

The source paper was identified through a public LinkedIn post. arXiv:2611.06237v1 and its Supporting Information were checked against data release 1.0.0 and public code commit 70707f5bb0c333ecec707e5a12b1efdf1c83603a. The code's MIT license was distinguished from the CC BY-NC 4.0 terms for the Tauri weights. The full datasets and model weights were not downloaded to run inference, training, or MD benchmarks. Property and speed figures from the paper are author-reported results.

OpenAI Codex root integrated the research, wrote the review, performed the educational LJ force, MD, and barrier-sensitivity calculations, and generated the cover. The mlff_data, mlff_physics, and mlff_benchmarks agents reviewed the data, physics, and benchmarks; intro_forcefield and mlff_taxonomy reviewed the expanded foundations and model explanations. The mlff_cover_brief agent supported cover planning, and review_format supported typesetting and output checks. The exact model identifier was not preserved. The review applied EDITORIAL_METHOD 2026.08 and the paragraph-context audit supplement of 1 November 2026. The cover is a conceptual illustration generated with the built-in image_gen tool; the four body figures in the original edition were created with code. LJ forces were checked by numerical differentiation, and MD was checked through energy error and time-step convergence. AI reviews covered the source text, numerical values, physics, models, explanations, and visual presentation. There is no record that a human completed a sentence-by-sentence review. For publication, mlff_publication_english provided the complete English translation and bilingual checks; mlff_publication_adapter produced the typeset mathematics, fonts and bilingual HTML/PDF; and mlff_publication_check independently reviewed the complete manuscript, paragraph context, bilingual correspondence and cover.

The 10 November 2026 revision rechecked the same arXiv v1 and data release 1.0.0, and statically inspected model registration, energy derivatives, and use terms at code commit ee62fa28860850d736845c3b91fba051d230f478. The original publication date remains 6 November. Root wrote the energy/chain-rule explanation, data-count clarification, observable-specific assessment, and new conceptual diagram. tauri_physics_update checked physics and official derivative paths; tauri_data_update checked datasets, training stages, and fixed versions; mlff_current_audit checked the baseline and build/date consistency. This revision did not rerun MD, training, or property benchmarks, and does not establish sentence-by-sentence final human review.